

PROGRAM 28:-

AIM

To implement the Forward Chaining technique in Prolog to derive new facts from existing facts and rules using suitable queries.

ALGORITHM

1. Start the program.
2. Define the initial facts in the knowledge base.
3. Define rules that derive new facts from existing facts.
4. Select the available facts and check the rules.
5. If the conditions of a rule are satisfied, derive a new fact.
6. Add the newly derived fact to the knowledge base.
7. Repeat the process until no new facts can be derived.
8. Execute queries to verify the derived facts.
9. Display the results of the queries.
10. Stop the program.

PROGRAM

```
fact(sunny).
```

```
fact(warm).
```

```
rule(good_weather, [sunny, warm]).
```

```
rule(go_outside, [good_weather]).
```

```
rule(play, [go_outside]).
```

```
forward :-
```

```
    fact(X),
```

```
    rule(Y, Conditions),
```

```
    \+ fact(Y),
```

```
    check(Conditions),
```

```
    assertz(fact(Y)),
```

```
    forward.
```

```
forward.  
check([]).  
check([H|T]) :-  
    fact(H),  
    check(T).
```

OUTPUT

```
53 ?- consult(['c:/Users/Hema Neepa/OneDrive/Documents/Forward_Chaining.pl']).  
Warning: c:/users/hema\_neepa/onedrive/documents/forward\_chaining.pl:8:  
Warning: Singleton variables: [X]  
true.  
  
54 ?- fact(good_weather).  
false.  
  
55 ?- fact(go_outside).  
false.  
  
56 ?- fact(play).  
false.
```

RESULT

The Forward Chaining technique was successfully implemented in Prolog, and the required queries were used to verify the facts derived from the given rules.